Aoji Tang

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Education

Lehigh University *Bethlehem, US*

PhD in Industrial Engineering *Aug 2026 - now*

- Advisor: Dr. Frank E. Curtis and Dr. Daniel P. Robinson

University of North Carolina at Chapel Hill (UNC) *Chapel Hill, US*

MS in Statistics, Exchange Program *Aug 2024 - Dec 2025, Fall 2023*

- Thesis advisor: Dr. Michael J. O'Neill
- Thesis title: An inexact two-stepsize stochastic SQP (sSQP) method for nonlinear optimization problems with rank-deficient constraint Jacobians

Fudan University *Shanghai, CN*

BS in Mathematics & Applied Mathematics *Sep 2019 - Jun 2024*

Publications

O'Neill, Michael J., and Aoji Tang. "Complexity of an inexact stochastic SQP algorithm for equality constrained optimization." arXiv preprint arXiv:2604.14351 (2026)

Research Experience

Spring2026 **Complexity Analysis of a Two-Phase SQP method,** Research Assistant *UNC*

- Designed a deterministic two-phase SQP method which yields a $\mathcal{O}(\epsilon_L^2 \log(\epsilon_c^{-1}))$ complexity.

Fall 2025 **Complexity Analysis on Constrained Optimization,** Research Assistant *UNC*

- Designed and analyzed an inexact two-stepsize sSQP algorithm for nonlinear constrained optimization under weak conditions.
- Derived first-known worst-case complexity bounds of sSQP method on constraint violation in the absence of constraint qualifications, whose order matches the existing optimum.
- Tested the algorithm on CUTEst and achieved lower infeasibility errors in less time than benchmark algorithms.

Fall 2024 **Universal Parameter-free Gradient-based Algorithm Analysis,** Individual *UNC*

- Extended the convergence result of Distance over Weighted Gradients (DoWG) algorithm to a strong stochastic gradient setting.
- Compared the performance of DoWG and three different types of gradient-based algorithms on deep learning tasks with Python.

Spring 2024 **Randomized Simplex Pivoting Rule,** Research Intern *UNC*

- Proposed and tested a new randomized pivoting rule of simplex method which avoids worst-case performance on Klee-Minty cubes with Matlab.
- Computed the expected pivoting times of the rule under a weighted discrete distribution.

Awards, Fellowships, & Grants

2025 **Master's Student Scholarship,** UNC STOR *$ 5,300*

Student Travel Grant, ICCOPT 2025 *$900*

Fall 2023 **Dean's List,** UNC (Exchange Program)

2023 **Professional Scholarship,** School of Mathematical Sciences, Fudan University *¥ 1,000*

2022 **Professional Scholarship,** School of Mathematical Sciences, Fudan University *¥ 1,000*

Teaching Experience

Fall 2025	**STOR 535: Probability for Data Science,** Instructional Assistant	*UNC*
Spring 2025	**STOR 155: Introduction to Data Models and Inference,** Teaching Assistant	*UNC*
Fall 2024	**STOR 415: Introduction to Optimization,** Teaching Assistant	*UNC*
Spring 2023	**Probability Theory and Mathematical Statistics,** Teaching Assistant	*Fudan*

Project & Workshop Experience

Fall 2024 **Reproducible Data Analysis on Enron Corpus Data,** Group Leader *UNC*

- Cleaned email data and clustered employees into nine business departments with R.
- Conducted semantic analysis with ChatGPT API on the email content.
- Maintained a (Nature Methods) silver-standard reproducible github repository. [link]

Fall 2024 **Tree Canopy Detection with Deep Learning,** Group Leader *UNC*

- Applied objection detection and image segmentation models such as YOLOv11, U-Net and SAM to detect and segment tree canopies for better managing forest resources.
- Labeled and annotated raw remote sensing images of forest to enlarge dataset. Enhanced shadow features to improve detection accuracy by 10%.

Aug 2023 **Polymath Jr. 2023,** Member *Online*

- Studied data assimilation, optimal transportation and their connection with generative machine learning with group members in weekly discussion.

Fall 2021 **Consumer Behavior Experiment and Analysis,** Group Leader *Fudan*

- Designed an offline social psychological experiment about consuming behavior, discounts and advertisements, and conducted it on 30 subjects.

Leadership & Extracurricular

2019-2021 **Fudan Art Troupe,** Director of Administration Dept. *Fudan*

- Supported business development of the university's graduation festival. Sponsorship added up to more than ¥180,000.
- Led a group of 20+ members on daily administrative work of the whole Troupe, managed to recruit more than 70 skilled new members.